

Functional genetic report · longevity

Male Fertility

Sérgio F · Report date: 02-10-2026

What the genes say about sperm production, its quality and the structural and tumor risks of the testis. Only genes (chip + local imputation + polygenic scores). Short report: the common genetics of male fertility is little known.

Contents

• Summary	3
• How to read this report	4
• Polygenic scores	5
• Spermatogenesis hormones (FSH, LH)	6
• Different channels (CFTR)	7
• Quality of sperm: methylation, antioxidants and structure	8
• Testis tumor	9
• Mechanisms of ageing	10
• Practical suggestions	11
• Limitations	12
• Methods and sources	13

Summary

16

SNPs analyzed

13

in the chip

2

imputed ($DR2 \geq 0.8$)

1

SNP not found in the data

2

Polygenic scores

Male infertility has strong but rare genetic causes (Y chromosome microdeletions, Klinefelter, CFTR variants, flagellum genes), which a chip practically does not read. The common part, read here, is made of small effects: FSH (FSHB, FSHR), methylation (MTHFR) and the antioxidant defense of spermatozoa. The testis tumor, on the contrary, has one of the greatest heritabilities of all cancers (KITLG, SPRY4, TERT).

In summary: no genetic signs of structural infertility visible on the chip and with genetic risk of low testis tumor (KITLG without the risk allele, which is the common one; score at the 24th percentile). The unfavourable point is FSHB -211G>T heterozygous, which decreases slightly the FSH and sperm production. On the side of drugs, rapamycin is the most relevant for fertility.

Favourable

- **KITLG rs995030 AA**: without the testis tumor risk allele, which ~80% of the reference population alleles have; testis tumor score at the 24th percentile.
- **CFTR R117H absent** and **FSHR N680/N680** (most sensitive to FSH).
- **FSHR -29 CC** (normal expression of the receptor) and LHCGR without the variant 312S.

Unfavourable

- **FSHB -211G>T heterozygous** (rs10835638 GT): FSH ~15% lower, testicular volume and sperm count slightly lower; FSH score at the 12th percentile.
- **MTHFR 677 and 1298 heterozygotes** (compound heterozygous) and GSTP1 Val105 heterozygous: small effects on sperm DNA quality.
- **Not visible**: CFTR F508del and 5T, AZF microdeletions of the Y chromosome and rare variants of the flagella.

Interpretation: the read genetics does not point to infertility. FSHB -211T is the only allele with known effect, small. The most relevant modifiable factor is pharmacological: rapamycin reduces spermatogenesis in a reversible manner.

How to read this report

Genetics only. All conclusions come only from Sérgio F genes (without blood tests or exams). They are predispositions, non-diagnosis. Each theme is divided into favourable and unfavourable; what does not have clear judgment appears as typical.

Strand: All genotypes are in the strand + of dbSNP (GRCh38). In the SNPs of this report, 13 directly agree with the dbSNP alleles, 0 needed conversion to the complementary strand and 0 are palindromic (A/T or C/G; orientation confirmed by the GRCh38 position). The genotypes are already in the strand + of the reference. Therefore, some genotypes appear with different letters from those used in the literature when using the strand of the gene (e.g. MTHFR C677T = rs1801133 A in the strand +).

Origin and trust: chip = directly genotyped (high confidence); imputation DR2=x = locally inferred from a public panel of complete genomes (30x), accepted only with $DR2 \geq 0.8$ (good trust, just below the chip); SNP not found in the data : neither chip nor imputed with sufficient quality (indels, rare variants, deletions).

Effect and judgment allele: The comparison between the number of Sérgio F copies and the mean of the reference population ($2 \times$ allele frequency; column «Freq.»). **Unfavourable** = at least 0.6 copies of risk allele above average (or any copy of a rare variant of strong effect); **Favourable** = at least 0.6 below average, protective allele, or absence of a strong effect variant; **Typical / no judgement** = close to the average (e.g. homozygous for a risk allele that is the most common) or context variant. Each common SNP has a small individual effect; what weighs more are the variants of large effect (marked as strong evidence) and the polygenic scores.

Polygenic scores (PGS): add thousands of variants with coefficients published in PGS Catalog. The percentile compares Sérgio F with the reference population (n=633), calculated with exactly the same variants. Percentage ≥ 80 in a risk score = unfavourable; ≤ 20 = favourable.

Polygenic scores

The testis tumor score (52 variants) was calculated in the cancer report and reused here. The simple FSH score uses only 5 variants of the GWAS Catalog and is at the 12th percentile (genetically low FSH, consistent with FSHB -211T). The simple male infertility score was not used: only 1 of 6 variants were available.

Trait	Score	Variables used (coverage)	Percentage (reference population)	Judgement
FSH (GWAS Catalog score, 5 loci)	C_FSH	5 of 5 (100%)	p12	Unfavourable
Testis tumor (52 variants)	PGS000086	48 of 52 (92%)	p24	Typical / no judgement

Spermatogenesis hormones (FSH, LH)

FSH stimulates Sertoli cells that support spermatogenesis; LH stimulates Leydig cells that produce testosterone within the testis.

Polygenic scores of this theme: FSH (GWAS Catalog score, 5 loci) (C_FSH): the 12th percentile Unfavourable

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
FSHB -211G>T	rs10835638	GT	chip	T	1	0.14	FSHB promoter: T allele reduces FSH (~15% per allele), testicular volume and sperm count; it is associated with male infertility. (strong evidence)	Unfavourable
FSHR N680S	rs6166	TT	chip	—	—	—	FSH receiver (Sertoli): S680 is slightly less sensitive to FSH (small effect on man). (moderate evidence)	Typical / no judgement
FSHR -29G>A	rs1394205	CC	chip	—	—	—	FSHR promoter: receptor expression. (weak evidence)	Typical / no judgement
LHCGR N312S	rs2293275	CC	chip	—	—	—	LH receiver: common polymorphism. (weak evidence)	Typical / no judgement

Favourable

- FSHR rs6166 TT (Asn680/Asn680)
- FSHR -29 CC
- LHCGR rs2293275 CC

Unfavourable

- FSHB -211 GT (1 allele T, ~14% of the reference population alleles)
- FSH score at the 12th percentile

Interpretation: The T allele of FSHB reduces the transcription of FSH. In heterozygous the effect is small and many men with this genotype are fertile. FSHR N680/N680 is the most sensitive form of the receiver, which partially compensates.

Different channels (CFTR)

CFTR variants in both alleles (e.g. F508del with the 5T variant) cause congenital absence of the deferens channels, a frequent cause of obstructive azoospermia.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
CFTR F508del	rs113993960		SNP not found in the data	—	—	—	CFTR: carriers of F508del + 5T/other variant may have congenital absence of the deferent channels (obstructive zoospermia). (strong evidence)	SNP not found in the data
CFTR R117H	rs78655421	GG	chip	A	0	—	CFTR R117H: typical variant of the agenesis of the different channels. (strong evidence)	Favourable
CFTR M470V	rs213950	GG	chip	—	—	—	CFTR M470V: modifies the channel function (relevant only with other variants). (weak evidence)	Typical / no judgement

Favourable

- CFTR R117H absent

Unfavourable

- F508del (deletion) and 5T (repetition) not visible in the chip

Interpretation: R117H and M470V read (without R117H; M470V GG, the common form). The most important CFTR variants for fertility are not read reliably by a chip; if there is a paternity project with difficulty, it is a directed test.

Quality of sperm: methylation, antioxidants and structure

The sperm DNA is very compacted (protamines) and very sensitive to oxidative stress; methylation (MTHFR) marks the paternal genome.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
MTHFR C677T	rs1801133	GA	chip	A	1	0.37	MTHFR 677T: associated in meta-analyses with lower and more fragmentation of sperm DNA. (moderate evidence)	Typical / no judgement
MTHFR A1298C	rs1801131	TG	chip	G	1	0.31	MTHFR 1298C: small effect. (weak evidence)	Typical / no judgement
SOD2 Val16Ala	rs4880	GG	chip	—	—	—	Mitochondrial SOD2: antioxidant defense of sperm. (weak evidence)	Typical / no judgement
GSTP1 Ile105Val	rs1695	AG	chip	G	1	0.34	GSTP1 Val105: associated with more oxidative stress in semen. (weak evidence)	Typical / no judgement
PRM1	rs737008	GT	imputation DR2=1.00	—	—	—	Protamine 1 (sperm DNA compaction): common polymorphism. (weak evidence)	Typical / no judgement
DNAH1	rs12163565	GG	imputation DR2=0.90	—	—	—	Belief dinein: rare variants cause immobile spermatozoa; this SNP is a common polymorphism. (weak evidence)	Typical / no judgement

Favourable

- PRM1 and DNAH1: common polymorphisms without effect

Unfavourable

- MTHFR 677 GA and 1298 TG
- GSTP1 Ile105Val AG

Interpretation: Small-effect findings, all heterozygous. SOD2 Ala16/Ala16: plus mitochondrial H2O2, which depends on normal catalase and GPX (see detoxification report).

Testis tumor

The testis germ cell tumor is the most frequent cancer in young men and has a strong genetic component.

Polygenic scores of this theme: Testis tumor (52 variants) (PGS000086): the 24th percentile

Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
KITLG	rs995030	AA	chip	G	0	0.80	KITLG: the strongest locus of the testis tumor (OR ~2.5 per allele). (strong evidence)	Favourable
SPRY4	rs4624820	GA	chip	A	1	0.53	SPRY4: testis tumor. (moderate evidence)	Typical / no judgement
TERT	rs2736100	CC	chip	—	—	—	TERT: testis tumor and telomeres. (moderate evidence)	Typical / no judgement

Favourable

- KITLG rs995030 AA (without risk allele, OR ~2.5 by allele)
- 52 variant score at the 24th percentile

Unfavourable

- SPRY4 rs4624820 GA (typical)

Interpretation: Low risk genetic profile. The peak incidence is before the 45 years.

Mechanisms of ageing

Each finding was linked to the aging mechanisms of the list of objectives for immortality (84 mechanisms). The table shows only the mechanisms played by this report.

Mechanism	Unfavourable	Favourable	Interpretation
Endocrine dysregulation	FSHB -211T	FSHR Asn680/Asn680; LHCGR	Mild weakness.
DNA Damage and DNA Repair Deficit	MTHFR heterozygous compound	Common PRM1	Small effect.
Oxidative stress	GSTP1 Val105; SOD2 Ala/Ala	—	Small effect.
Epigenetic Changes	MTHFR 677/1298	—	Methylation of the paternal genome.
Somatic Mutation Accumulation	—	KITLG without risk; p24 testis tumor score	Favourable.
Dysfunction of Primary Cyls	—	Common DNAH1 (flagellum)	Neutral.

Practical suggestions

All the suggestions below are suggestion (not applied): only from genetics, they are aligned with the v2 genetic plan and do not change the current stack without Sérgio F decision and the doctor.

Topic	Suggestion (not applied)	Genetic basis
Rapamycin and fertility	If there is a plan to have children, cryopreserve semen before long cycles of rapamycin, or pause for 3 months (the duration of spermatogenesis) with the doctor; the effect of rapamycin on spermatogenesis is reversible. suggestion (not applied)	FSHB -211T (minor margin)
Testicular heat	With regular hammock, sperm count is temporarily low; if there is a paternity plan, pause the 3 swarm months earlier. suggestion (not applied)	FSHB -211T
Folate and B12	Natural folate (vegetable, legumes) and B12 sufficient; if supplemental folate, prefer methylfolate. suggestion (not applied)	MTHFR heterozygous compound
Antioxidants of semen	Fruit and vegetables, nuts (omega-3 vegetable) and fish oil that already takes (3.2-4 g/day); zinc and selenium for food. suggestion (not applied)	GSTP1 Val105; SOD2 Ala/Ala
Exposures	Not heating food in plastic and reducing pesticides (washing well the fruit and vegetables; biological in the most contaminated). suggestion (not applied)	Oxidative stress
Testis self-examination	Monthly: the genetic risk is low, but the exam is simple and the tumor has a very high cure when caught early. suggestion (not applied)	P24 score
Coordination with the plan	D3 10 000 IU and omega-3 3.2-4 g/day as in the plan. No telomerase activators. Curcumin, luteolin, apigenin and quercetin in pause while ALT is high. suggestion (not applied)	V2 plan rules

Limitations

- The strong genetic causes of male infertility (AZF microdeletions of Y, karyotype, F508del and 5T of CFTR, rare genes of flagellum and meiosis) are not read by the chip.
- The GWAS of male fertility are small; the male infertility score did not have enough variants and the FSH uses only 5 loci.
- The percentiles compare with the reference population (n=633).
- Only genes: does not use spermogram or blood analysis.

Methods and sources

- Genotypes: Raw DNA of the chip (GRCh38, strand +), analyzed only locally; Missing SNPs (statistical imputation, public reference panel of complete 30x genomes), accepted with $DR2 \geq 0.8$.
- Public positions and alleles: Ensembl REST (only rsIDs). Locally calculated frequencies in the reference population ($n=633$); effect alleles confirmed by the expected frequency.
- Scores: PGS000086 (testis tumor, 52 variants; reused from the cancer report) and a simple GWAS Catalog score (local copy) for FSH.
- Main references: Grigorova 2008 and Tüttelmann 2012 (FSHB -211G>T); Simoni 2002 (FSHR N680S); Chillón 1995 (CFTR and agenesis of the different channels); Gupta 2011 (MTHFR and infertility, meta-analysis); Kanetsky 2009 and Rapley 2009 (KITLG, SPRY4); Pickard 2019 (TGCT); Huyghe 2007 (mTOR inhibitors and male gonads); Garolla 2013 (heat and semen).
- Mapping for 84 aging mechanisms of the target list. Report generated locally (HTML → PDF).